Practice Problems
Calculus-Based Physics II

Chapter 37 Electric Field and Conductors

1) A positive charge of 4 μC is placed on a solid conducting sphere, what happens to the charge on the outer surface of the sphere.

   A) The charge stays where it was placed.
   B) The charge moves throughout the volume of the sphere.
   C) The charge moves to the surface and is uniformly distributed.
   D) Depends on the type of charge.

2) A positive charge of 4 μC is placed on a solid conducting sphere, what happens to the electric field inside the sphere.

   A) The electric field strength is created by the net charge.
   B) The electric field strength is pointing out from the center.
   C) The electric field strength is pointing in towards the center.
   D) The electric field strength is zero.

3) A positive charge of 4 μC is placed on a hollow conducting sphere, what happens to the electric field inside the sphere's empty cavity.

   A) The electric field strength is created by the net charge.
   B) The electric field strength is pointing out from the center.
   C) The electric field strength is pointing in towards the center.
   D) The electric field strength is zero.

4) The electric field line emanating from a conduction surface are

   A) Always perpendicular to the conduction surface.
   B) Always parallel to the conducting surface.
   C) Oriented along the conducting surface.
   D) At random angles to the conducting surface.

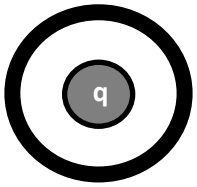
5) A solid aluminum cube rests on a wooden table in a region where a uniform external electric field is directed straight upward. What can we say concerning the charge on the **top surface** of the cube?

   A) The top surface is charged positively.
   B) The top surface is charged negatively.
   C) The top surface is neutral.
   D) The top surface's charge cannot be determined without further information.

6) A solid aluminum cube rests on a wooden table in a region where a uniform external electric field is directed straight upward. What can we say about the electric field inside the cube?

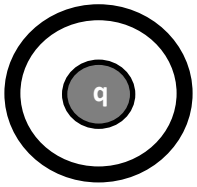
   A) It is equal to the external electric field.
   B) It is equal to zero.
   C) It is larger than the external electric field.
   D) It is smaller than the external electric field but not zero.

7) A small conducting sphere with a net charge of positive q is placed inside a larger neutral conducting sphere, they are not touching, what is the charge on the outside of the larger sphere when the charge has time to equalizes?



A) Negative q
B) Positive q
C) No charge, it is neutral

8) A small conducting sphere with a net charge of positive q is placed inside a larger neutral conducting sphere, they are not touching, what is the charge on the inside of the larger sphere when the charge has time to equalize?



A) Negative q
B) Positive q
C) No charge, it is neutral

9) What type of charge does a cloud have when lighting strikes the ground?

A) Positive charge.
B) Negative charge.
C) Either positive or negative charge.
D) No charge.

10) When a car is struck by lightning, the resulting electric field inside the car is

A) zero.
B) normally huge for a time longer than the lightning stroke itself.
C) small enough to be safe for an occupant inside.
D) normally huge, but for a brief time.

Answers

1) C
2) D
3) D
4) A
5) A
6) B
7) B
8) A
9) C
10) A